



Analysis and Design of Algorithms



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رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي وَاحْلُلْ عُقْدَةً مِنْ لِسَانِي يَفْقَهُوا قَوْلِي



O my Lord! Expand for me my chest [with assurance] and ease for me my task and untie the knot from my tongue that they may understand my speech. (Quran 20 : 25-28)



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Algorithm Analysis

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Introduction

- Algorithm Analysis
- Methodology
 - Empirical Analysis
 - **Experimental** based analysis
 - Mathematical / Theoretical Analysis
 - **Mathematical** based analysis

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Empirical Analysis

Method 01

w.r.t. Problem Size and Hardware Variations



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Empirical Analysis of Time Efficiency

- Select multiple hardware setup
- Select a specific (typical) sample of inputs
- Use physical unit of time (e.g., milliseconds) or Count actual number of basic operation's executions
- Analyze the empirical data

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Empirical Analysis

Type of Computer	Time (sec)
Home Computer	51.915
Desktop Computer	11.508
Minicomputer	2.382
Mainframe Computer	0.431
Supercomputer	0.087
Time to sort an array of 2000 integers	

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Empirical Analysis

Array Size	Home Comp	Desktop Comp
125	12.5	2.8
250	49.3	11.0
500	195.8	43.4
1000	780.3	172.9
2000	3114.9	690.5

Insertion Sort Running Time for Different Input Size

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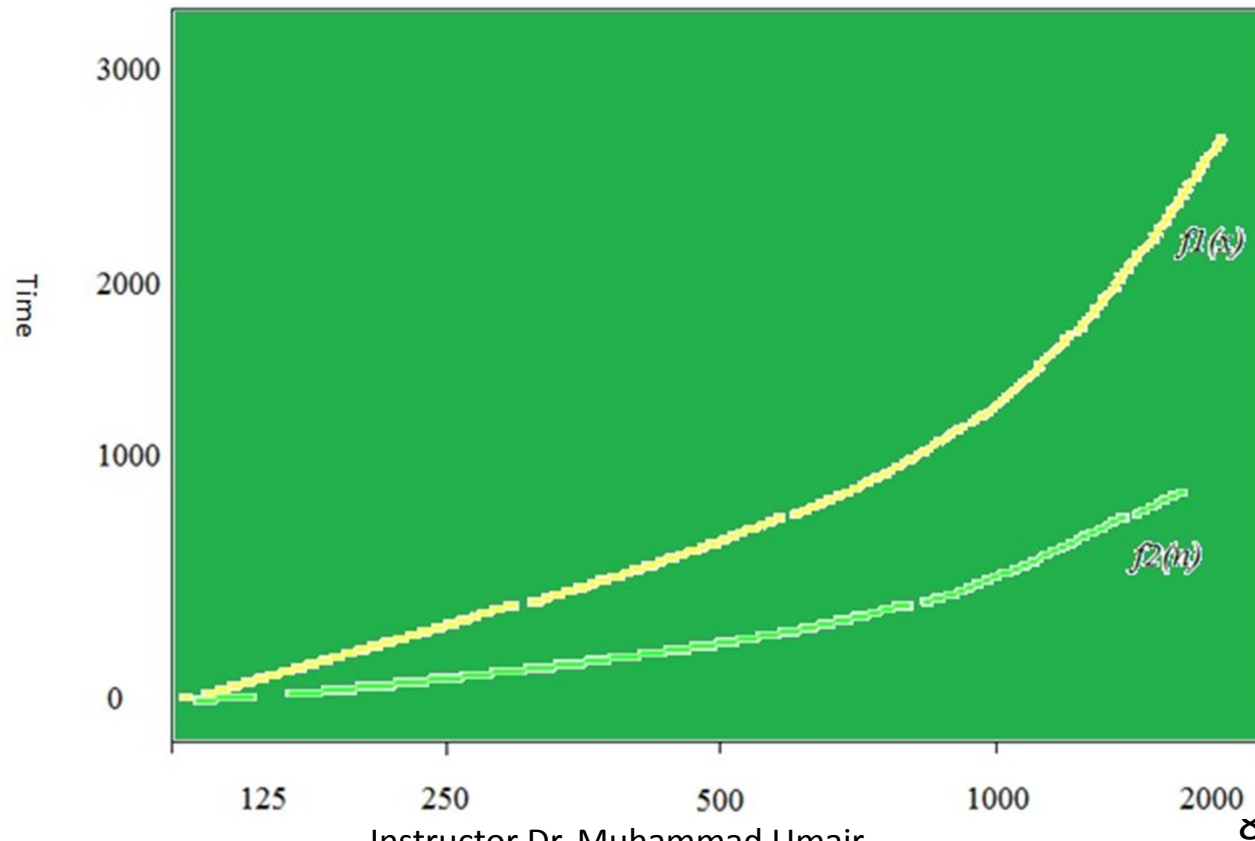
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Empirical Analysis

$$f1(n) = 0.0007772 n^2 + 0.00305 n + 0.001$$

$$f2(n) = 0.0001724 n^2 + 0.00040 n + 0.100$$



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Empirical Analysis

n	f(n)	$a n^2$	% of $a n^2$
125	2.8	2.7	96.4
250	11.0	10.8	98.1
500	43.4	43.1	99.1
1000	172.9	172.4	99.7
2000	690.5	689.6	99.9

$$f(n) = a n^2 + b n + c$$

where $a = 0.0001724$, $b = 0.0004$ and $c = 0.1$

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Empirical Analysis

Method 02

W.r.t. Problem Size



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General Plan

- Understand the experiment's **purpose**
- Decide on the **efficiency metric M** to be measured and the **measurement unit** (an operation count-using counter vs. a time unit-microsecond or nanosecond)
- Decide on **characteristics of the input** sample (its range, size, and so on)
- Prepare a **program implementing** the algorithm (or algorithms) for the experimentation
- **Generate a sample** of inputs
- Run the algorithm (or algorithms) on the sample's inputs and **record the data** observed
- **Analyze** the data obtained



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General Plan

- Understand the experiment's **purpose**
 - **Checking the accuracy** of theoretical analysis results
 - **Comparison of efficiency** of different algorithms of same problem
 - **Developing the hypothesis** about the algorithm's efficiency
 - **Ascertaining the efficiency** of the algorithm on a particular machine

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General Plan

- Decide on the **efficiency metric M** to be measured and the **measurement unit** (an operation count vs. a time unit)
 - **Operation Count**
 - insert a **counter** (or counters) into a program implementation
 - Count number of times the basic operation is implemented
 - Basic Operation may be executed in several places in the program
 - **Time Measurement**
 - **Use stopwatch** to measure the time taken to complete the task
 - **Use a system's command**, such as the time command in UNIX
 - measure the running time of a code fragment by asking for **the system time** right before the fragment's start (t_{start}) and just after its completion (t_{finish}). The difference between the two ($t_{\text{finish}} - t_{\text{start}}$)
 - In Java, the method `currentTimeMillis()` in the `System` class is available

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General Plan

- Decide on **characteristics of the input** sample (its range, size, and so on)
 - start with a **relatively small** sample
 - **increase it later** if necessary
 - **range** of instance sizes; neither **trivially small** nor **excessively large**

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General Plan

- Prepare a **program implementing** the algorithm (or algorithms) for the experimentation
 - Select any **suitable high-level programming language**
 - C, C++, Java etc.
 - May be implemented in **multiple languages**
 - Observe the consistency in the running time

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General Plan

- **Generate a sample** of inputs
 - empirical analysis requires **generating random numbers**
 - random number generator available in computer **language libraries**
 - value of a (pseudo)random variable uniformly distributed in the interval **between 0 and 1**
 - an **appropriate transformation** may be used for desired range of value
 - if **x** is a continuous **random variable** uniformly distributed on the interval $0 \leq x < 1$
 - **$y = l + \text{floor}(x(r - l))$**
`Random rand = new Random();`
`int min=50; max=100;`
`System.out.println(rand.nextInt(max - min + 1) + min);`
 - Generates the value between integers **l and r - 1**
 - » where $(l < r)$



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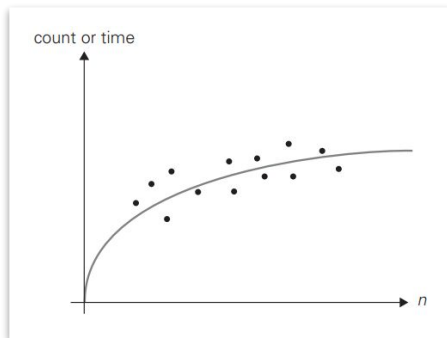
General Plan

- Run the algorithm (or algorithms) on the sample's inputs and **record the data** observed
 - **Using Table**
 - Numerical data will be presented
 - Manipulated easily
 - Ratio $M(n) / g(n)$ e.g. $M(2n) / M(n)$
 - Change slightly **Logarithmic**
 - Converge to 2 **Linear**
 - Converge to 4 **Quadratic**
 - Converge to 8 **Cubic**
 - **Graphical – Scatterplot**
 - Draw points in a Cartesian coordinate systems
 - Concave **Logarithmic**
 - Points aggregation around line **Linear**
 - Convex Shape **$n \lg n$ or n^2**
 - Convex with rapid increase **n^3**

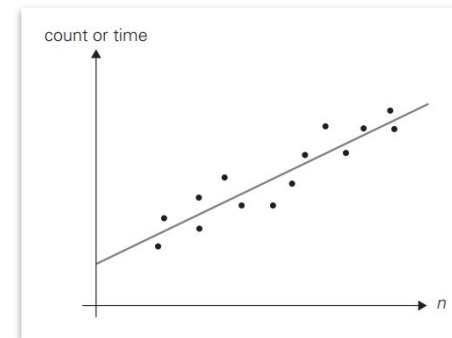
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General Plan

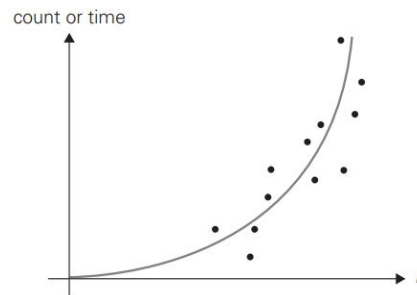
- Run the algorithm (or algorithms) on the sample's inputs and **record the data** observed



Logarithmic



Linear



Quadratic



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Algorithm Visualization

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Introduction

- The use of images to convey some useful information about algorithms
 - **Standard Simulation Tools**
 - Microsoft Excel
 - **Customized Application**
 - High Level Language, Visual Tool etc.
 - Algorithm visualization was introduced first in 1970 by Ronald Baecker & D. Sherman (University of Toronto)
 - **Presented 30-minute** color sound film titled ***Sorting Out Sorting***
 - It contained **visualization** of **09 well-known sorting algorithms**

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Methods & Tools

- Methods
 - **Static** Algorithm Visualization
 - shows an algorithm's progress through a series of still images
 - **Dynamic** Algorithm Visualization / Algorithm Animation
 - shows a continuous, movie-like presentation of an algorithm's operations
- Tools
 - **Bar graph**
 - via vertical or horizontal bars or sticks of different heights or lengths
 - Rearranged according to their sizes
 - Useful for small inputs
 - **Scatterplot**
 - Useful for larger files
 - Presenting points on a coordinate plane
 - first coordinate representing an item's position in the file
 - second one representing the item's value

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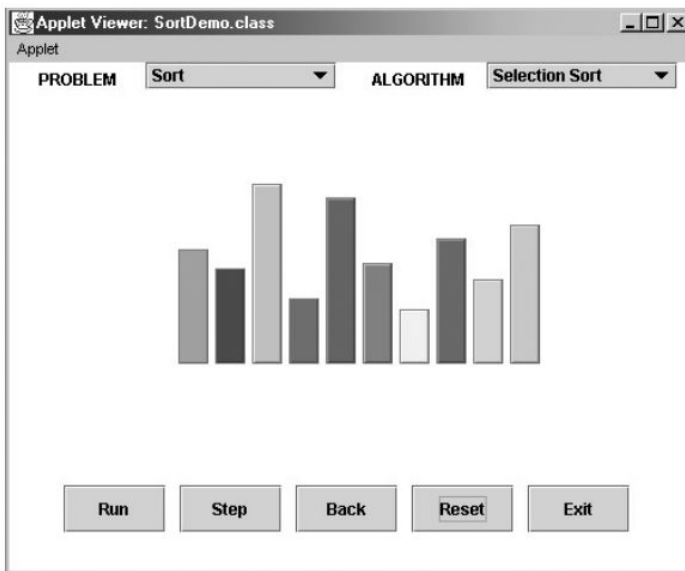
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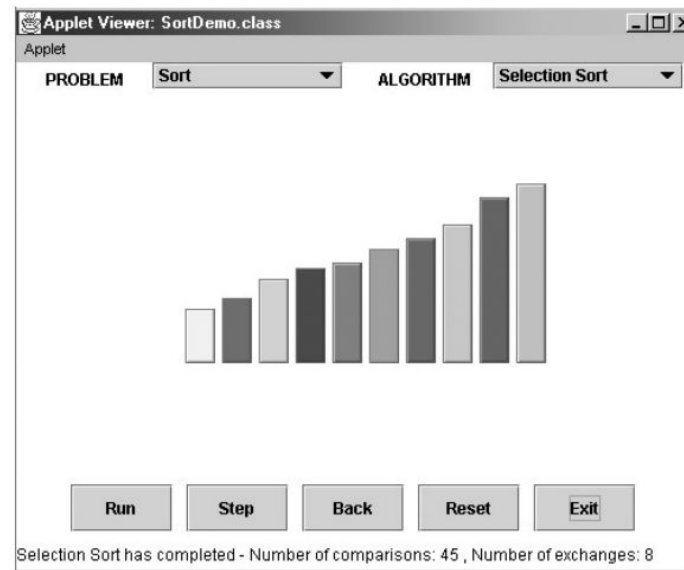
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Bar Graph



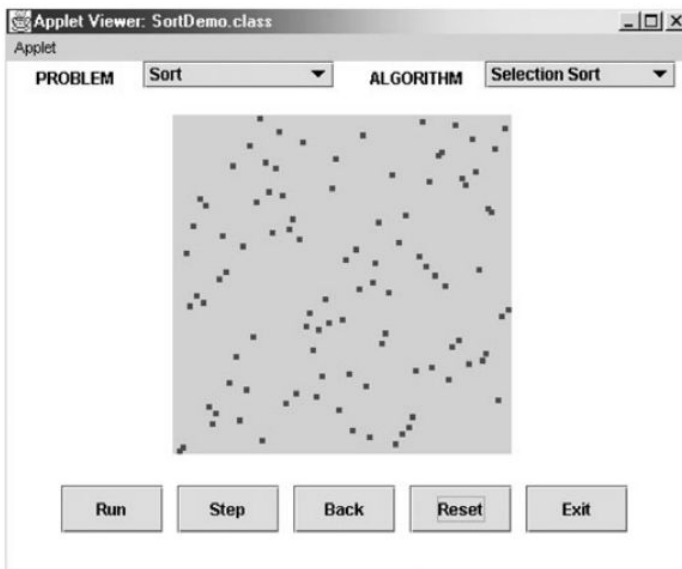
Before Sorting



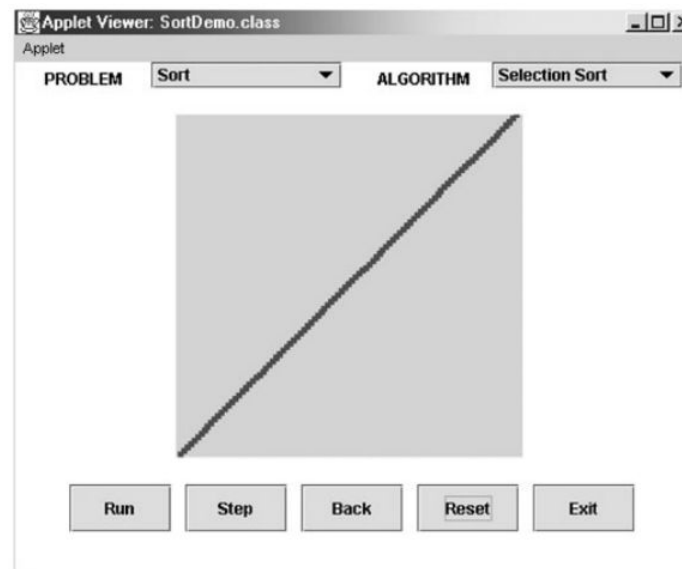
After Sorting

<https://visualgo.net/en/sorting?slide=1>

Scatterplot



Before Sorting



After Sorting

<https://www.youtube.com/watch?v=DSMCZZGbZo4>



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Allah (Alone) is Sufficient for us, and He is the Best Disposer of affairs (for us). (Quran 3 : 173)